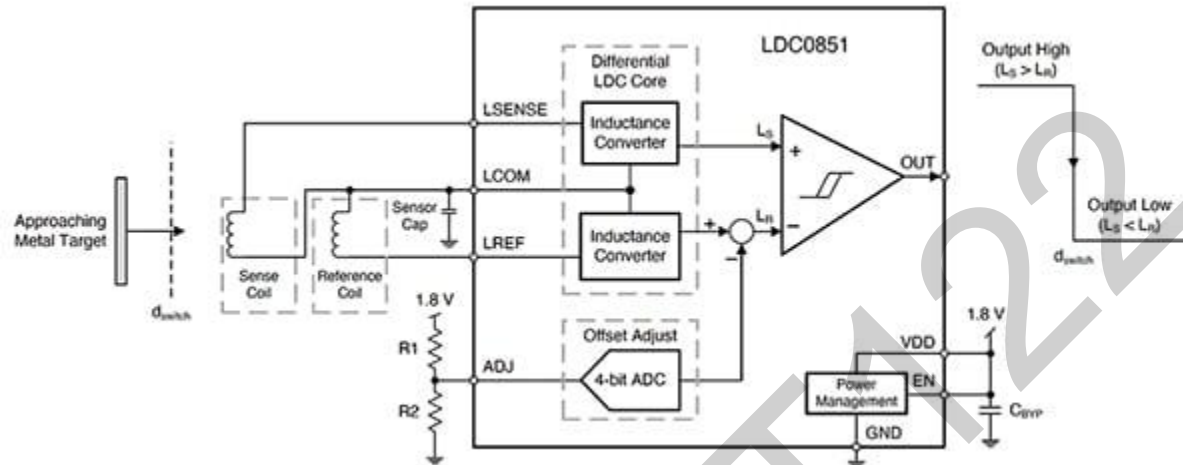


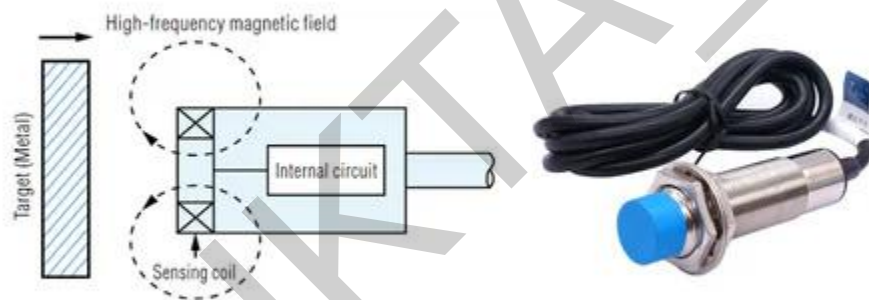
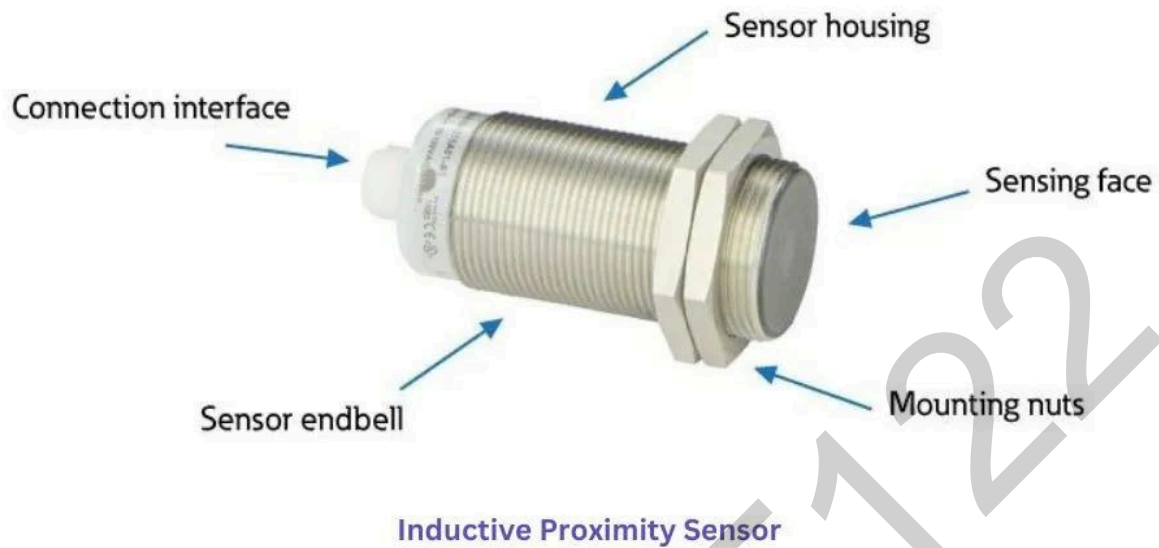
## Practical 1 - Study of Sensor Node Hardware Architecture

### Block Diagram :



### Definition

A Wireless Sensor Node is a compact electronic device that senses physical parameters, processes the collected data, stores it temporarily, and transmits it wirelessly to a base station or monitoring system. According to the uploaded PDF, a WSN consists of sensors, sensor nodes, a base station, and a graphical user interface for monitoring and control.



## Application

### Industrial Conveyor Belt Metal Object Detection

An inductive proximity sensor is installed beside a conveyor belt in a manufacturing plant. As metallic products pass near the sensor, it detects their presence without physical contact. The wireless sensor node processes the detection signal and sends the information wirelessly to the control room for monitoring production, counting products, or detecting missing items.

## Role of Each Component in an Energy-Constrained Environment

| Component                                    | Role                                                                                                                                                                                      | Energy-Efficient Operation                                                                                                                     |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Sensing Unit<br>(Inductive Proximity Sensor) | Detects nearby metallic objects by generating a high-frequency electromagnetic field. The presence of metal changes the inductance or oscillation frequency, indicating object detection. | Activated only when monitoring is required to reduce power consumption.                                                                        |
| Microcontroller (MCU)                        | Receives sensor signals, processes detection data, filters noise, counts objects, and decides when to transmit information.                                                               | Uses sleep mode between sensing intervals and wakes only when the sensor detects a metal object.                                               |
| Memory                                       | Stores the program code, sensor calibration values, temporary detection counts, and communication data.                                                                                   | Reduces repeated sensing and communication by buffering data before transmission.                                                              |
| Wireless Transceiver                         | Sends processed data to the base station using ZigBee, LoRa, or Wi-Fi and receives control commands if needed.                                                                            | Since wireless communication consumes the most energy, the transceiver remains in sleep mode and transmits only when significant events occur. |
| Power Supply                                 | Supplies energy to all components using batteries, often supported by power management circuits or energy harvesting.                                                                     | Efficient voltage regulation and duty cycling maximize battery life, enabling long-term operation without frequent battery replacement.        |

## Working of the Inductive Proximity Sensor

1. According to the provided information:
2. The sensor generates a high-frequency electromagnetic field using an oscillating coil.
3. When a metal object enters this field, eddy currents are induced in the object.
4. These eddy currents create an opposing magnetic field, reducing the sensor's inductance or changing its oscillation frequency.

5. The sensor detects this change and produces an output signal.
6. The microcontroller processes the signal and transmits the information wirelessly to the monitoring station.

### Why It Is Suitable for Energy-Constrained WSNs

1. Non-contact sensing minimizes mechanical wear.
2. The sensor operates only when metal objects are nearby.
3. The microcontroller spends most of its time in low-power sleep mode.
4. Data is transmitted only after object detection, reducing wireless communication energy.
5. Battery-powered operation allows deployment in remote industrial environments without continuous power supply.

### Conclusion

An inductive proximity sensor-based wireless sensor node is highly suitable for industrial automation, such as conveyor belt metal object detection, event counting, machine monitoring, and production line automation. The sensing unit, microcontroller, memory, transceiver, and power supply work together while employing sleep modes, event-driven sensing, and minimized wireless transmissions to maximize battery life in energy-constrained environments. This aligns with the WSN architecture described in your PDF, where sensor nodes collect data and transmit it wirelessly to a base station for monitoring and control.

#### 1. Battery Level vs Time

| Time  | Battery (%) |
|-------|-------------|
| 09:00 | 100         |
| 10:00 | 98          |
| 11:00 | 96          |

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|       |    |
|-------|----|
| 12:00 | 94 |
| 01:00 | 92 |
| 02:00 | 90 |

### 2. Detection Events per Hour

| Hour  | Objects Detected |
|-------|------------------|
| 9 AM  | 150              |
| 10 AM | 180              |
| 11 AM | 210              |
| 12 PM | 190              |
| 1 PM  | 220              |

**Graph :**

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## Metal Objects Detected Over Time

Number of metal objects detected by the inductive proximity sensor during conveyor operation.

